

Lab 3 - Routing Protocol

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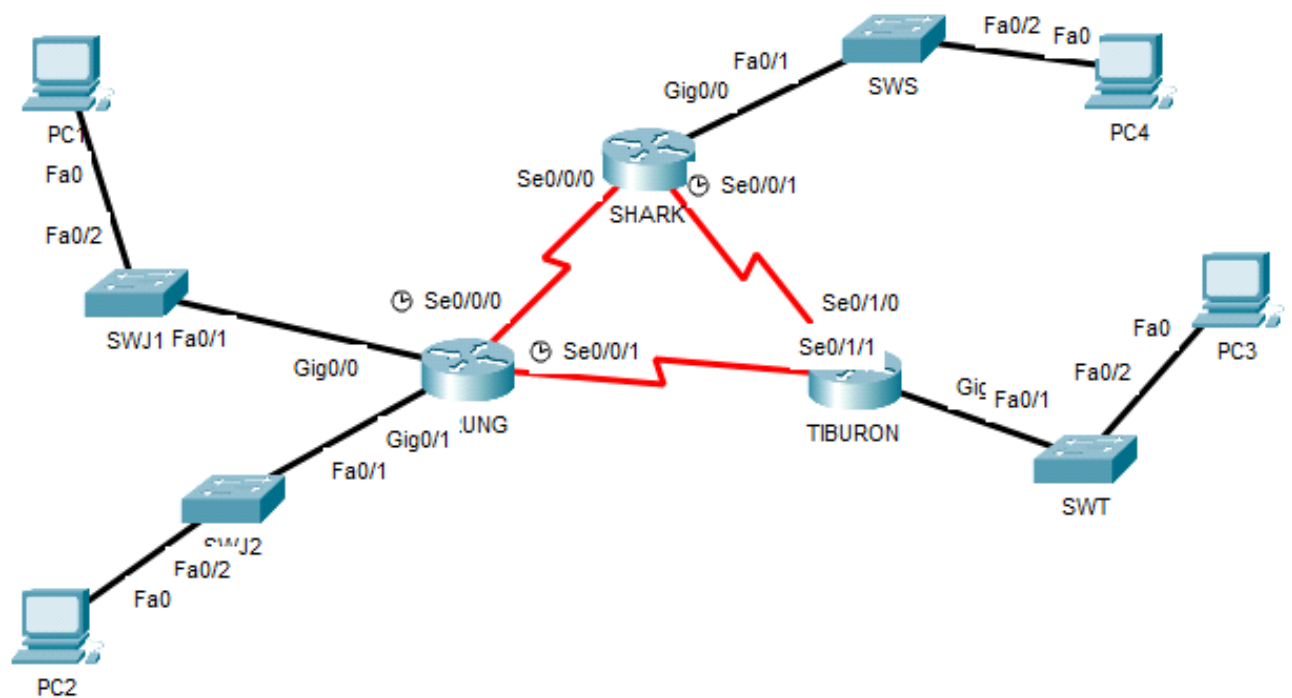
A23CS0207

SECTION: 02

INTRODUCTION

This lab looks into the network layer, focusing on subnetting and routing.

TOPOLOGY



LAB INFORMATION

Here is some basic information for the lab. The network address given is 172.18.110.0/23.

Table 1

Device	Subnetwork	Usable Hosts
JERUNG	LAN1	20
	LAN2	30
SHARK	LAN	60
TIBURON	LAN	50
Connections	JERUNG-SHARK	2
	JERUNG-TIBURON	2
	SHARK-TIBURON	2

Table 2

#	Device Name	Interface	IP Address	Subnet Mask	Gateway
1	JERUNG	Se0/0/0	172.18.110.193	255.255.255.252	
2		Se0/0/1	172.18.110.197	255.255.255.252	
3		G0/0	172.18.110.27	255.255.255.224	172.18.110.27
4		G0/1	172.18.110.43	255.255.255.224	172.18.110.43
5	TIBURON	Se0/1/1	172.18.110.198	255.255.255.252	
6		Se0/1/0	172.18.110.202	255.255.255.252	
7		G0/0	172.18.110.66	255.255.255.192	172.18.110.66
8	SHARK	Se0/0/0	172.18.110.194	255.255.255.252	
9		Se0/0/1	172.18.110.201	255.255.255.252	
10		G0/0	172.18.110.123	255.255.255.192	172.18.110.123
11	PC1	-	172.18.110.28	255.255.255.224	172.18.110.27
12	PC2	-	172.18.110.58	255.255.255.224	172.18.110.43
13	PC3	-	172.18.110.120	255.255.255.192	172.18.110.66

14	PC4	-	172.18.110.182	255.255.255.192	172.18.110.123
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LAB TASKS

Task 1 – IP Addressing

- Given the network address of the organisation and the basic information provided in both Tables 1 and 2. Show your workings here and complete Table 2 with the correct information. PCs will be given the last usable address of the subnetwork.
- Using the IP addresses calculated, configure the devices with the appropriate information.

Forung G0/0 LAN1

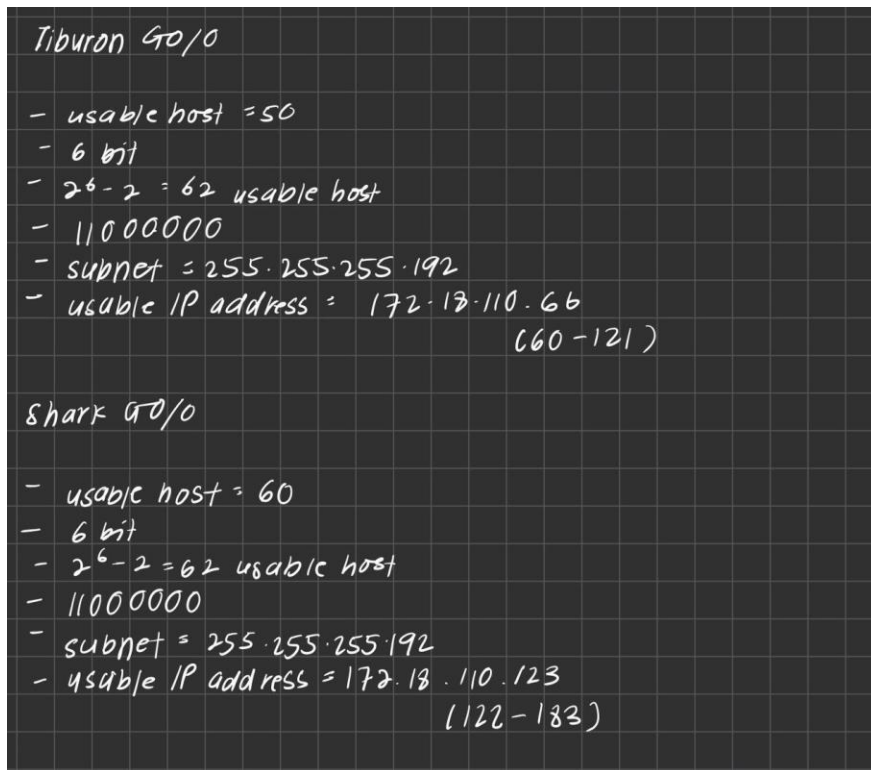
- usable host = 20
 - 5 bit
 - $2^5 - 2 = 30$ usable host
 - 11100000
 - subnet = 255.255.255.224
 - usable IP address = 172.18.110.27/27
 (0-29)

11111111.11111111.11111111.11100000

network host

Forung G0/1 LAN2

- usable host = 30
 - 5 bit
 - $2^5 - 2 = 30$ usable host
 - 11100000
 - subnet = 255.255.255.224
 - usable IP address = 172.18.110.43
 (30-59)



Task 2 – Routing Table

- Paste the current routing table of each router here. There are 2 ways to this – via CLI and via Packet Tracer (PT) tool. Use both ways to show the routing table of all the routers.
 - CLI
 - Click on a router. In the prompt, type show ip route. The routing table of the router will be shown, as shown in Figure A.

```
SHARK#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    172.18.0.0/16 is variably subnetted, 2 subnets, 2 masks
C       172.18.110.0/26 is directly connected, GigabitEthernet0/0
L       172.18.110.1/32 is directly connected, GigabitEthernet0/0
```

```

SHARK>enable
SHARK#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    172.18.0.0/16 is variably subnetted, 6 subnets, 3 masks
C       172.18.110.64/26 is directly connected, GigabitEthernet0/0
L       172.18.110.123/32 is directly connected, GigabitEthernet0/0
C       172.18.110.192/30 is directly connected, Serial0/0/0
L       172.18.110.194/32 is directly connected, Serial0/0/0
C       172.18.110.200/30 is directly connected, Serial0/0/1
L       172.18.110.201/32 is directly connected, Serial0/0/1

```

CLI SHARK

```

JERUNG>enable
JERUNG#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    172.18.0.0/16 is variably subnetted, 8 subnets, 3 masks
C       172.18.110.0/27 is directly connected, GigabitEthernet0/0
L       172.18.110.27/32 is directly connected, GigabitEthernet0/0
C       172.18.110.32/27 is directly connected, GigabitEthernet0/1
L       172.18.110.43/32 is directly connected, GigabitEthernet0/1
C       172.18.110.192/30 is directly connected, Serial0/0/0
L       172.18.110.193/32 is directly connected, Serial0/0/0
C       172.18.110.196/30 is directly connected, Serial0/0/1
L       172.18.110.197/32 is directly connected, Serial0/0/1

```

CLI JERUNG

```

TIBURON>enable
TIBURON#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    172.18.0.0/16 is variably subnetted, 6 subnets, 3 masks
C       172.18.110.64/26 is directly connected, GigabitEthernet0/0
L       172.18.110.66/32 is directly connected, GigabitEthernet0/0
C       172.18.110.196/30 is directly connected, Serial0/1/1
L       172.18.110.198/32 is directly connected, Serial0/1/1
C       172.18.110.200/30 is directly connected, Serial0/1/0
L       172.18.110.202/32 is directly connected, Serial0/1/0

```

CLI TIBURON

Figure A

- PT tool
 - Click on the magnifying glass (shown in Figure B), then click on a router, then choose Routing Table. A sample of the result is shown in Figure C.

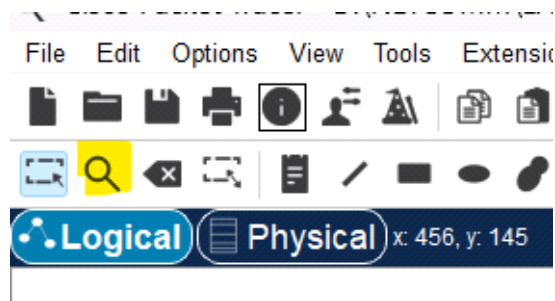


Figure B

Routing Table for SHARK					
Type	Network	Port	Next Hop IP	Metric	
C	172.18.110.0/26	GigabitEthernet0/0	---	0/0	
L	172.18.110.1/32	GigabitEthernet0/0	---	0/0	

Figure C

Routing Table for SHARK					
Type	Network	Port	Next Hop IP	Metric	
C	172.18.110.64/26	GigabitEthernet0/0	---	0/0	
L	172.18.110.123/32	GigabitEthernet0/0	---	0/0	
C	172.18.110.192/30	Serial0/0/0	---	0/0	
L	172.18.110.194/32	Serial0/0/0	---	0/0	
C	172.18.110.200/30	Serial0/0/1	---	0/0	
L	172.18.110.201/32	Serial0/0/1	---	0/0	

PT TOOL SHARK

Routing Table for JERUNG				
Type	Network	Port	Next Hop IP	Metric
C	172.18.110.0/27	GigabitEthernet0/0	---	0/0
L	172.18.110.27/32	GigabitEthernet0/0	---	0/0
C	172.18.110.32/27	GigabitEthernet0/1	---	0/0
L	172.18.110.43/32	GigabitEthernet0/1	---	0/0
C	172.18.110.192/30	Serial0/0/0	---	0/0
L	172.18.110.193/32	Serial0/0/0	---	0/0
C	172.18.110.196/30	Serial0/0/1	---	0/0
L	172.18.110.197/32	Serial0/0/1	---	0/0

PT TOOL JERUNG

Routing Table for TIBURON				
Type	Network	Port	Next Hop IP	Metric
C	172.18.110.64/26	GigabitEthernet0/0	---	0/0
L	172.18.110.66/32	GigabitEthernet0/0	---	0/0
C	172.18.110.196/30	Serial0/1/1	---	0/0
L	172.18.110.198/32	Serial0/1/1	---	0/0
C	172.18.110.200/30	Serial0/1/0	---	0/0
L	172.18.110.202/32	Serial0/1/0	---	0/0

PT TOOL TIBURON

- Try to ping PC2 from PC1, paste the results here.


```

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 172.18.110.58

Pinging 172.18.110.58 with 32 bytes of data:

Request timed out.
Reply from 172.18.110.58: bytes=32 time=1ms TTL=127
Reply from 172.18.110.58: bytes=32 time=1ms TTL=127
Reply from 172.18.110.58: bytes=32 time<1ms TTL=127

Ping statistics for 172.18.110.58:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>|

```

- Try to ping PC4 from PC1, paste the results here.

```

C:\>ping 172.18.110.182

Pinging 172.18.110.182 with 32 bytes of data:

Reply from 172.18.110.27: Destination host unreachable.
Reply from 172.18.110.27: Destination host unreachable.
Reply from 172.18.110.27: Destination host unreachable.
Reply from 172.18.110.27: Destination host unreachable.

Ping statistics for 172.18.110.182:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>

```

- Explain the reason(s) behind the results.

From the first diagram which is ping PC2 from PC1, it has 25% loss because out of 4 packets sent, only 1 packet loss. This is because, they only share a router. For PC4 and PC1, it has 100% loss because out of 4 packets sent, all packets loss. This is because, they don't share the same router.

- What needs to be done to ensure all PCs can ping each other successfully?

Configure correct IP address, subnet masks, and default gateways for each PC. Then, make sure all router interfaces and devices are functional. After that, ping between PCs to test the connectivity between them.

Task 2 – Routing Configuration

- Let's start by opening the routing table (using the PT tool) for TIBURON and JERUNG. This is done to show changes to the routing table as configurations are made.

Routing Table for TIBURON						Routing Table for JERUNG					
Type	Network	Port	Next Hop IP	Metric		Type	Network	Port	Next Hop IP	Metric	
C	172.18.110.64/26	GigabitEthernet0/0	---	0/0		C	172.18.110.128/27	GigabitEthernet0/1	---	0/0	
L	172.18.110.65/32	GigabitEthernet0/0	---	0/0		L	172.18.110.129/32	GigabitEthernet0/1	---	0/0	
C	172.18.110.196/30	Serial0/1/1	---	0/0		C	172.18.110.160/27	GigabitEthernet0/0	---	0/0	
L	172.18.110.198/32	Serial0/1/1	---	0/0		L	172.18.110.161/32	GigabitEthernet0/0	---	0/0	

Figure D

- In router JERUNG, configure the RIP routing protocol as shown in Figure E.

```

JERUNG#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
JERUNG(config)#router rip
JERUNG(config-router)#version 2
JERUNG(config-router)#network 172.18.110.128
JERUNG(config-router)#network 172.18.110.160
JERUNG(config-router)#network 172.18.110.192
JERUNG(config-router)#network 172.18.110.196
JERUNG(config-router)#no auto-summary
JERUNG(config-router)#

```

Figure E

- What can you say about the addresses used in the 'network' instructions in Figure E?

The address used in the 'network' represent subnets are directly connected to the JERUNG router by using the RIP.

- Then configure RIP in TIBURON. All are similar except use the network address. Use the instructions as shown below.

```

TIBURON (config-router) #network 172.18.110.64
TIBURON (config-router) #network 172.18.110.196
TIBURON (config-router) #network 172.18.110.200

```

- As you may have seen, there are changes in the routing tables of both TIBURON and JERUNG. Paste a copy of these routing tables here.

Routing Table for JERUNG					Routing Table for TIBURON				
Type	Network	Port	Next Hop IP	Metric	Type	Network	Port	Next Hop IP	Metric
C	172.18.110.0/27	GigabitEthernet0/0	---	0/0	R	172.18.110.0/27	Serial0/1/1	172.1...	120/1
L	172.18.110.27/32	GigabitEthernet0/0	---	0/0	R	172.18.110.32/27	Serial0/1/1	172.1...	120/1
C	172.18.110.32/27	GigabitEthernet0/1	---	0/0	C	172.18.110.64/26	GigabitEthernet0/0	---	0/0
L	172.18.110.43/32	GigabitEthernet0/1	---	0/0	L	172.18.110.66/32	GigabitEthernet0/0	---	0/0
R	172.18.110.64/26	Serial0/0/1	172.1...	120/1	R	172.18.110.192/30	Serial0/1/1	172.1...	120/1
C	172.18.110.192/30	Serial0/0/0	---	0/0	C	172.18.110.196/30	Serial0/1/1	---	0/0
L	172.18.110.193/32	Serial0/0/0	---	0/0	L	172.18.110.198/32	Serial0/1/1	---	0/0
C	172.18.110.196/30	Serial0/0/1	---	0/0	C	172.18.110.200/30	Serial0/1/0	---	0/0
L	172.18.110.197/32	Serial0/0/1	---	0/0	L	172.18.110.202/32	Serial0/1/0	---	0/0
R	172.18.110.200/30	Serial0/0/1	172.1...	120/1					

ROUTING TABLE JERUNG AND TIBURON

- What are the changes seen in TIBURON?

TIBURON's routing table in Figure D only contains directly connected (C) and local (L) routes. No RIP (R) routes are present in this state. Meanwhile, new RIP (R) routes appear in the TIBURON's routing table.

- What are the Networks with type R in TIBURON and JERUNG?

JERUNG:

- 172.18.110.64/26
- 172.18.110.200/30

TIBURON

- 172.18.110.0/27
 - 172.18.110.32/27
 - 172.18.110.192/30
- Ping PC3 from PC1. Was it successful?
No.
 - Ping PC4 from PC1. Was it successful?
No.

- Explain the reasons for your answer.

PC3 from PC1 got 25% packet loss, meanwhile PC4 from PC1 got 100% packet loss.

- Continue with configuration of RIP in SHARK. Paste your configurations here.

```
SHARK>enable
SHARK#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
SHARK(config)#router rip
SHARK(config-router)#version 2
SHARK(config-router)#network 172.18.110.193
SHARK(config-router)#network 172.18.110.197
SHARK(config-router)#network 172.18.110.27
SHARK(config-router)#network 172.18.110.43
SHARK(config-router)#network 172.18.110.198
SHARK(config-router)#network 172.18.110.202
SHARK(config-router)#network 172.18.110.66
SHARK(config-router)#network 172.18.110.194
SHARK(config-router)#network 172.18.110.201
SHARK(config-router)#network 172.18.110.123
SHARK(config-router)#network 172.18.110.28
SHARK(config-router)#netork 172.18.110.58
      ^
% Invalid input detected at '^' marker.

SHARK(config-router)#network 172.18.110.58
SHARK(config-router)#network 172.18.110.120
SHARK(config-router)#network 172.18.110.182
SHARK(config-router)#no auto-summary
SHARK(config-router)#
```

- Open router SHARK's routing table and paste here.

Routing Table for SHARK

Type	Network	Port	Next Hop IP	Metric
R	172.18.110.0/27	Serial0/0/0	172.18.110.193	120/1
R	172.18.110.32/27	Serial0/0/0	172.18.110.193	120/1
C	172.18.110.64/26	GigabitEthernet0/0	---	0/0
L	172.18.110.123/32	GigabitEthernet0/0	---	0/0
C	172.18.110.192/30	Serial0/0/0	---	0/0
L	172.18.110.194/32	Serial0/0/0	---	0/0
R	172.18.110.196/30	Serial0/0/0	172.18.110.193	120/1
R	172.18.110.196/30	Serial0/0/1	172.18.110.202	120/1
C	172.18.110.200/30	Serial0/0/1	---	0/0
L	172.18.110.201/32	Serial0/0/1	---	0/0

- Try to ping from PC4 to all other PCs in the topology. *Note: Try to ping at least twice to get best results.

PC	Ping result
1	0% packet loss
2	0% packet loss
3	0% packet loss

Task 3 – Routing Update

- Let's try a little experiment. Change the IP addresses of router TIBURON interface G0/0 to 192.168.1.1/24.
 - This means that the subnet has changed. Find the new Network address of this subnet. Network Address is: 192.168.1.0
 - As this change happens, PC3 must also have different IP address, subnet mask and gateway address. What will it be?

PC3	Info
IP address	192.168.1.60
Subnet Mask	255.255.255.0

Gateway Address	192.168.1.1
-----------------	-------------

- After this change, can PC4 and PC1 ping PC3?

Yes. PC4 can ping PC3.

- Copy and paste the routing tables for both SHARK and TIBURON here.

Routing Table for SHARK					Routing Table for TIBURON				
Type	Network	Port	Next Hop IP	Metric	Type	Network	Port	Next Hop IP	Metric
R	172.18.110.0/27	Serial0/0/0	172.18.110.193	120/1	R	172.18.110.0/27	Serial0/1/1	172.18.110.197	120/1
R	172.18.110.32/27	Serial0/0/0	172.18.110.193	120/1	R	172.18.110.32/27	Serial0/1/1	172.18.110.197	120/1
C	172.18.110.64/26	GigabitEthernet0/0	---	0/0	R	172.18.110.64/26	Serial0/1/0	172.18.110.201	120/1
L	172.18.110.123/32	GigabitEthernet0/0	---	0/0	R	172.18.110.192/30	Serial0/1/1	172.18.110.197	120/1
C	172.18.110.192/30	Serial0/0/0	---	0/0	R	172.18.110.192/30	Serial0/1/0	172.18.110.201	120/1
L	172.18.110.194/32	Serial0/0/0	---	0/0	C	172.18.110.196/30	Serial0/1/1	---	0/0
R	172.18.110.196/30	Serial0/0/0	172.18.110.193	120/1	L	172.18.110.198/32	Serial0/1/1	---	0/0
R	172.18.110.196/30	Serial0/0/1	172.18.110.202	120/1	C	172.18.110.200/30	Serial0/1/0	---	0/0
C	172.18.110.200/30	Serial0/0/1	---	0/0	L	172.18.110.202/32	Serial0/1/0	---	0/0
L	172.18.110.201/32	Serial0/0/1	---	0/0	C	192.168.1.0/24	GigabitEthernet0/0	---	0/0
					L	192.168.1.1/32	GigabitEthernet0/0	---	0/0

- Referring to the routing table, explain your findings.

The routing table for SHARK and TIBURON shows that both routers are directly connected to the networks.

- What is your next move to ensure end-to-end connectivity (i.e. all PCs can ping each other successfully)?

Configure routing protocol so that all PC's can exchange routing information and make all networks reachable.

- Show your configurations in TIBURON to ensure end-to-end connectivity.

```

TIBURON>enable
TIBURON#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
TIBURON(config)#router tip
      ^
% Invalid input detected at '^' marker.

TIBURON(config)#router rip
TIBURON(config-router)#version 2
TIBURON(config-router)#network 192.168.1.0
TIBURON(config-router)#

```

- To ensure end-to-end connectivity, ping to all the PCs from PC3.

PC	Ping result
1	<pre> Cisco Packet Tracer PC Command Line 1.0 C:\>ping 172.18.110.28 Pinging 172.18.110.28 with 32 bytes of data: Reply from 172.18.110.28: bytes=32 time=1ms TTL=126 Reply from 172.18.110.28: bytes=32 time=1ms TTL=126 Reply from 172.18.110.28: bytes=32 time=1ms TTL=126 Reply from 172.18.110.28: bytes=32 time=5ms TTL=126 Ping statistics for 172.18.110.28: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 1ms, Maximum = 5ms, Average = 2ms C:\> </pre>
2	<pre> C:\>ping 172.18.110.58 Pinging 172.18.110.58 with 32 bytes of data: Reply from 172.18.110.58: bytes=32 time=9ms TTL=126 Reply from 172.18.110.58: bytes=32 time=4ms TTL=126 Reply from 172.18.110.58: bytes=32 time=4ms TTL=126 Reply from 172.18.110.58: bytes=32 time=1ms TTL=126 Ping statistics for 172.18.110.58: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), Approximate round trip times in milli-seconds: Minimum = 1ms, Maximum = 9ms, Average = 4ms C:\> </pre>
4	<pre> C:\> ping 172.18.110.182 Pinging 172.18.110.182 with 32 bytes of data: Reply from 192.168.1.1: Destination host unreachable. Reply from 192.168.1.1: Destination host unreachable. Reply from 192.168.1.1: Destination host unreachable. Reply from 192.168.1.1: Destination host unreachable. Ping statistics for 172.18.110.182: Packets: Sent = 4, Received = 0, Lost = 4 (100% loss), C:\> </pre>

REFLECTION

What have you learned in this task?

I learned a lot from this lab session and improved my knowledge of routing and subnetting. Assigning IP addresses, setting up routers, and resolving connectivity problems made me realize how important precise setups are to efficient network communication. Resolving packet loss between PCs taught me the value of functional router interfaces, default gateways, and appropriate subnet masks. My understanding of how networks exchange data to maintain connectivity has improved as a result of setting up the RIP protocol and seeing how it affects

routing tables. All things considered, this lab helped me gain a practical understanding of the problems and solutions associated with network infrastructure management.